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Bulk and metastable defects in $\text{CuIn}_{1-x}\text{Ga}_x\text{Se}_2$ thin films using drive-level capacitance profiling

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The drive-level capacitance profiling technique has been applied to $\text{ZnO}/\text{CdS}/\text{CuIn}_{1-x}\text{Ga}_x\text{Se}_2/\text{Mo}$ solar cell devices, in order to study properties of defects in the $\text{CuIn}_{1-x}\text{Ga}_x\text{Se}_2$ film. Properties studied include the spatial uniformity, bulk defect response, carrier density, and light-induced metastable effects. These results indicate that previous estimates of carrier densities, from $C-V$ profiling, may be significantly overestimated. In addition, a defect response previously thought to be located at the interface is observed to exist throughout the bulk material. Finally, an infrared light-soaking treatment is demonstrated to induce metastable changes in the bulk $\text{CuIn}_{1-x}\text{Ga}_x\text{Se}_2$ film. Hence, the drive-level capacitance profiling technique provides valuable insights into these films. Herein, the technique itself is fully explained, compared to other junction capacitance methods, and its utility is demonstrated using numerical simulation. © 2004 American Institute of Physics. [DOI: 10.1063/1.1633982]

I. INTRODUCTION

It has long been recognized that measurements of junction capacitance can yield valuable insights into the density of defect states in the bandgap of semiconductor materials, as well as other electronic properties such as the density of free carriers. Techniques based on junction capacitance investigate the semiconductor in its working environment, as a part of a simple device (usually a one-sided $p-n$ junction or a Schottky diode). This is particularly important in the study of materials whose electronic properties may be affected by their working environment within the device. Such environmental conditions include electric fields, such as that provided by the built-in potential, as well as surface and interface conditions. Polycrystalline $\text{CuIn}_{1-x}\text{Ga}_x\text{Se}_2$ (CIGS) thin films are an example of one such class of sensitive materials.

The chalcopyrite CIGS materials have been largely studied because of their application as the absorber layer in thin-film solar cells, due to their near-optimal, and tuneable (by variation of x), bandgap. In the course of investigating these materials, it has become clear that they have interesting electronic properties. For example, the completed devices undergo metastable changes due to light exposure and applied bias (see, e.g., Refs. 1–4), and this appears to result at least partially from changes in the defect density and carrier density in the film. This could be due to migration of copper ions, which are apparently highly mobile.^{5–7} Under some conditions, light exposure has even been found to improve the efficiency of the completed devices. In order to begin to understand these and other effects, it is crucial that we be able to accurately determine such basic properties as the net

free carrier density in the film, and the density of defect states in the bandgap.

A number of junction capacitance techniques have been developed to probe such basic properties. However, one must be cautious in interpreting measurements of junction capacitance for semiconductor materials that may have significant densities of defect states within the bandgap. Such materials include amorphous and polycrystalline thin films. For example, in these films, the standard interpretation of measurements of junction capacitance as a function of applied dc bias ($C-V$) can significantly overestimate the carrier density.

We have found that drive-level capacitance profiling (DLCP) is a useful technique for studying amorphous and polycrystalline semiconductors. Indeed, it was developed with the study of these materials in mind.⁸ This technique directly yields the density of states within the bandgap of the semiconductor, as a function of both energy and of spatial position, requiring knowledge only of the semiconductor dielectric constant (ϵ). It also provides a much more accurate assessment of the free carrier density in the film than the $C-V$ method. In addition, DLCP measurements are primarily sensitive to *bulk* states; that is, they are not generally skewed by the presence of interface states. Finally, this technique is not difficult to apply.

We have recently found the DLCP technique to be particularly insightful in the study of polycrystalline CIGS thin films. In prior studies of these films using admittance spectroscopy, a predominant defect response is observed at energies varying between 0.1 and 0.3 eV, depending on the device studied. This response has been christened “ N_1 ,” and it has been suggested that it originates from the $p-n$ junction interface.⁹ Since N_1 seems to be one of the more universally

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observed defect-state transitions in these films, its nature and its effects need to be better understood.

We will illustrate why DLCP is a valuable technique for this endeavor; however, we also note that this technique has been previously explained in the literature only briefly. Hence, we will also undertake to describe the DLCP technique in some detail, and compare it to other, more commonly employed junction admittance techniques. We begin in Sec. II by reviewing the general principle behind two related junction capacitance techniques: C - V and admittance spectroscopy. In Sec. III, we present a theoretical development of the DLCP technique, and comment on the experimental requirements. Next, in Sec. IV, numerical simulations of the C - V and DLCP experiments are presented for certain scenarios that may occur in CIGS films. These simulations demonstrate the aforementioned difficulties in obtaining free carrier densities from C - V measurements on certain materials, as well as the effects of interface states on these measurements. After a brief discussion of our instrumentation and sample series in Sec. V, we then discuss, in Sec. VI, the results of some of our DLCP experiments on CIGS devices. Detailed comparisons of actual DLCP and C - V measurements reinforce our concern that carrier densities obtained from C - V measurements may be substantially overestimated. Most significantly, the DLCP technique has allowed us to determine that N_1 is a true bulk defect in our CIGS films. We are also able to deduce its density and spatial uniformity. Finally, we demonstrate that its density can be altered in a metastable fashion using a light exposure treatment. Sec. VII summarizes these results.

II. BACKGROUND

One of the most common junction capacitance techniques is a simple C - V measurement at room temperature. In this measurement, an ac voltage perturbation is applied to the device, and the quadrature current response is measured, to obtain the junction capacitance. The variation in junction capacitance with dc bias is then determined. In the standard interpretation, these measurements give the charge density at the edge of the depletion region [$N(W)$]:

$$N(W) = -\frac{C^3}{q\epsilon A^2(dC/dV)} = -\frac{2}{q\epsilon A^2} \left[\frac{d(1/C^2)}{dV} \right]^{-1}, \quad (1)$$

where W is width of the depletion region, A is the area of the device, and q is the fundamental charge.¹⁰ This assumes a one-sided device, in which the dominant capacitance response originates from the edge of the depletion region in the more intrinsic layer, and assumes that the temperature is high enough that the carriers can be rapidly transported into and away from the edge of the depletion region at the applied ac frequency. The derivation of Eq. (1) explicitly assumes that, as the dc bias is changed, the charge density only changes at the edge of the depletion region. For a semiconductor in which one has close to 100% ionization of the shallow dopants, $N(W)$ is generally assumed to correspond to the density of free carriers.

In films with significant densities of states in the depletion region, however, there can be an additional dynamic

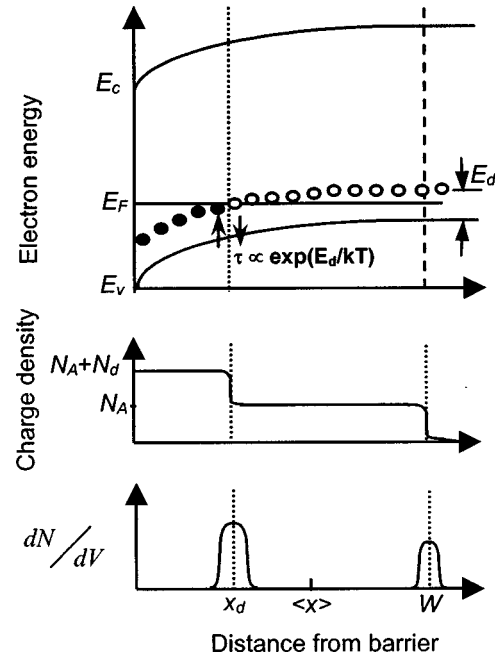


FIG. 1. Response of a sample with a single discrete defect, illustrated for the p -side of a $n^+ - p$ junction.

response to the applied ac bias at the positions where the electronic levels of the defect cross the quasi-Fermi level within the depletion region. Due to this additional response, the junction capacitance response can be generally written as

$$C = \frac{\epsilon A}{\langle x \rangle}, \quad (2)$$

where $\langle x \rangle$ is the first moment of charge response:

$$\langle x \rangle = \frac{\int_0^\infty x \delta\rho(x) dx}{\int_0^\infty \delta\rho(x) dx}. \quad (3)$$

In this equation, x is the distance from the barrier location and $\delta\rho(x)$ is the variation of charge density across the semiconductor caused by the oscillating voltage. Thus, the standard interpretation of the C - V technique breaks down when $\langle x \rangle$ is significantly shifted from the edge of the depletion region W , due to significant densities of states deeper within the energy gap.

Such a scenario is illustrated in Fig. 1 for a $n^+ - p$ -type junction. The p -type film has a shallow dopant density N_A , a single defect with density N_d , and activation energy E_d as measured from the valence band edge. In this illustration, the applied ac voltage induces a change in charge in the film both at the edge of the depletion region and at the position where the defect crosses E_F . This will occur as long as the ac angular frequency, ω , is slow enough that the response time of the defect $\tau < 1/\omega$. The first moment of charge response is then $\langle x \rangle = (x_d N_d + W N_A) / (N_d + N_A)$.

The C - V technique is also less accurate for semiconductor films that have significant densities of interface states. These situations have been carefully analyzed previously; for example, see Crowell and Roberts.¹¹ The C - V technique can be greatly skewed by such complications since they affect the manner in which the junction capacitance varies during

the relatively long time between changes in the dc bias. Additionally, any such slowly responding deep defects or interface states will affect the C - V measurement in a manner that can vary dramatically depending upon the rate at which the dc bias is changed.

Another standard junction capacitance measurement is the measurement of junction capacitance as a function of frequency and temperature, known as admittance spectroscopy.¹² This technique can yield the activation energy of conduction, as well as the activation energy of any defect states having energetic positions that cross the quasi-Fermi energy within the band-bending region. Generally, admittance spectroscopy measurements are sensitive to states with energies between that of the quasi-Fermi energy at the interface, and the activation energy of conduction. While the results from admittance spectroscopy can be distorted by spatially nonuniform densities of defects, measurements at a number of different dc biases can test whether such spatial nonuniformity is a concern. In addition, since response from interface states only occurs when their energetic position intersects that of the quasi-Fermi level near the interface, measurements at more than one dc bias can often distinguish whether a particular response arises from interface or bulk states.¹³

While admittance spectroscopy readily indicates the activation energy of the defect response, it is more difficult to obtain the absolute density of such states from this measurement since such estimates require prior knowledge of other material characteristics.¹⁴ This can be a problem particularly in cases in which the defect density becomes comparable to the carrier density, due to the dependence of these calculations on estimates of the carrier density, generally obtained by the C - V technique.

Therefore, C - V and admittance spectroscopy techniques are valuable in the study of semiconductor materials, provided they do not have a large density of deep levels, significant nonuniformities, large interface densities of states, or other anomalies. Thus, they may be able to provide only a very crude estimate of the properties of polycrystalline and amorphous materials.

III. THEORY OF DRIVE-LEVEL CAPACITANCE PROFILING

Drive-level capacitance profiling (DLCP) is an alternate technique for measuring the energetic and spatial distribution of defects in the bandgap.⁸ The DLCP method examines the junction capacitance dependence over a range of ac voltage amplitudes in addition to a range of dc bias, to obtain more information about the film response than is available from the aforementioned C - V and admittance spectroscopy techniques. This technique has previously been employed to study the density of states in amorphous silicon thin films, and has greatly contributed to understanding the electronic properties of that material.¹⁵⁻¹⁷ Here, we illustrate how it can also be very useful in studying CIGS films.

In standard admittance techniques, the capacitance is measured for only very small ac voltages, and hence the charge response is assumed to be linear with the applied

voltage: $dQ = CdV$. In DLCP, larger ac biases are used, and the additional nonlinear terms become important:

$$dQ/dV = C_0 + C_1 dV + C_2 (dV)^2 + \dots \quad (4)$$

The DLCP technique takes advantage of the information embedded in these higher order terms to deduce the defect density N_d . This can be understood by looking more closely at the capacitance response of a depletion region containing a continuous density of gap states.¹⁸

The (angular) frequency ω and temperature T of the measurement determine the limiting energy E_e at which the occupation of a gap state can change rapidly enough to follow the applied alternating bias:

$$E_e = -kT \ln \left(\frac{\omega}{2\pi\nu_0 T^2} \right). \quad (5)$$

Here, $\nu = \nu_0 T^2 = N_v(T) \langle v \rangle \sigma_h$ is the thermal emission prefactor, and reflects the effective density of states in the valence band N_v , the average thermal velocity $\langle v \rangle$, and the capture cross section of the gap state involved (σ_h). Since changes in state occupation can only follow the alternating voltage for states closer to the valence band edge than E_e (for a p -type material), only these states influence the measured value of junction capacitance at a given ω and T .

The position x_e is defined to be the location at which $E_F - E_V = E_e$, and is determined from the band bending in the depletion region. Between x_e and the interface, the occupation of any gap states crossing the quasi-Fermi level do not usually have time to follow the ac voltage, since in this region, $E_F - E_V > E_e$. At distances larger than x_e , gap states lying at energies between E_F and E_e will respond at frequency ω . This situation is illustrated in Fig. 2.

In general, the dc potential at the interface, $\psi_0(x=0)$, can be expressed as

$$\begin{aligned} \psi_0(0) = & \int_0^\infty x \frac{\rho_0(x)}{\epsilon} dx = \int_0^{x_e} x \frac{\rho_0(x)}{\epsilon} dx \\ & + \int_{x_e}^\infty x \frac{\rho_0(x)}{\epsilon} dx, \end{aligned} \quad (6)$$

where $\rho(x)$ is the charge density in the absence of the applied ac voltage. Assuming a highly doped and thus very thin depleted layer on the n side of the junction, adding a small voltage (δV) will increase the value of the potential at the barrier interface to $\psi(0) = \psi_0(0) + \delta V$. The position at which $E_F - E_V = E_e$ will then move farther from the barrier interface, to $x_e + \delta x$, where x_e still labels the position where $E_F - E_V = E_e$ prior to adding δV (see Fig. 2). We will denote the charge density distribution after δV is added as $\rho(x)$.

If δV is applied only for a time of order $1/\omega$, then the gap states lying between $x=0$ and $x=x_e$ will not have sufficient time to change their occupation. In that case, $\rho(x)$ will remain equal to $\rho_0(x)$ in this part of the depletion region. Moreover, in the region between x_e and $x_e + \delta x$, the occupation of states will be *emission limited* so that the charge density will be roughly constant and equal to the charge density we previously had at the location x_e prior to

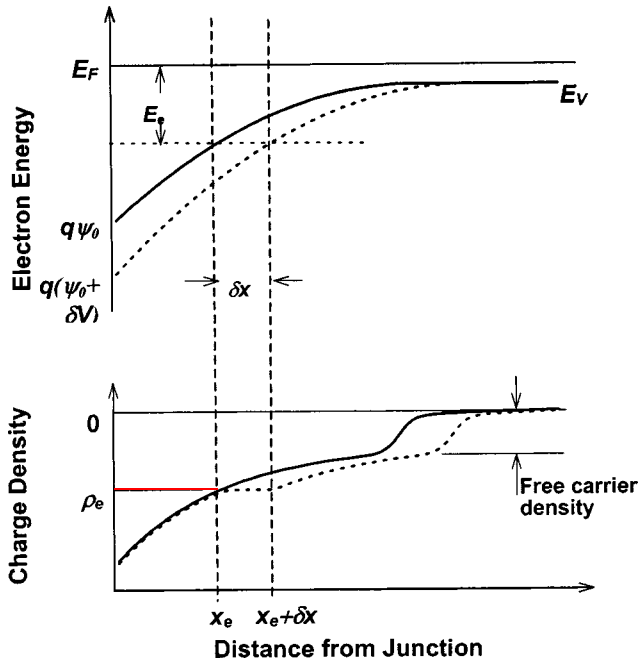


FIG. 2. Upper curves indicate schematically the change in band bending when a small bias (δV) is applied: the solid line and dashed lines indicate the band bending before and after the application of δV , respectively. Lower curves show the corresponding charge densities as solid and dashed lines. The time scale of the measurement determines E_e and x_e , such that the trapped charge has time to equilibrate only at positions where $E_F - E_V < E_e$.

the application of δV . We shall denote the value of this charge density as ρ_e . Therefore, at a time $1/\omega$ following the application of δV , we will have

$$\begin{aligned} \psi(0) = & \int_0^{x_e} x \frac{\rho_0(x)}{\epsilon} dx + \int_{x_e}^{x_e + \delta x} x \frac{\rho_e}{\epsilon} dx \\ & + \int_{x_e + \delta x}^{\infty} x \frac{\rho(x)}{\epsilon} dx. \end{aligned} \quad (7)$$

As long as beyond the point x_e the properties of the semiconductor do not vary rapidly on a length scale of δx , then the final integral in Eq. (7) will be equal to the final integral in Eq. (6). This is because both integrals encompass depletion regions having exactly the same boundary conditions with gap state occupations in quasi-equilibrium. Specifically, we will have $\rho(x) \approx \rho_0(x - \delta x)$. Therefore, δQ , the total change in depletion region charge in response to δV , will be simply the charge lying between x_e and $x_e + \delta x$, or

$$\delta Q = A \rho_e \delta x. \quad (8)$$

If the voltage δV is now removed, the initial charge distribution will be restored within a time also of order $1/\omega$. Thus, the capacitance due to an oscillating voltage that toggles the junction potential between ψ_0 and $\psi_0 + \delta V$ at frequency ω will simply be $\delta Q / \delta V$, with δQ given by Eq. (8).

The remaining task in obtaining our final expression for this capacitance will be to express δx in terms of δV . This follows from Eq. (7). Since $\rho(x) \approx \rho_0(x - \delta x)$ and ρ_e is a constant, we have

$$\begin{aligned} \psi(0) = & \int_0^{x_e} x \frac{\rho_0(x)}{\epsilon} dx + \frac{\rho_e}{2\epsilon} [(x_e + \delta x)^2 - x_e^2] \\ & + \int_{x_e + \delta x}^{\infty} x \frac{\rho_0(x - \delta x)}{\epsilon} dx, \end{aligned} \quad (9)$$

so that

$$\begin{aligned} \delta V = & \psi(0) - \psi_0(0) \\ = & \frac{\rho_e}{2\epsilon} [(x_e + \delta x)^2 - x_e^2] + \int_{x_e}^{\infty} \delta x \frac{d^2 \psi}{dx^2} dx, \end{aligned}$$

or

$$\delta V = \frac{\rho_e}{2\epsilon} (2x_e \delta x + \delta x^2) - \delta x F_e,$$

where

$$F_e = \left| \frac{d\psi}{dx} \right|_e$$

is the magnitude of the electric field at the point x_e .

Solving for δx :

$$\delta x = \left(\frac{\epsilon}{\rho_e} F_e - x_e \right) \left[1 - \sqrt{1 + \frac{2\rho_e \epsilon \delta V}{(\epsilon F_e - \rho_e x_e)^2}} \right]. \quad (10)$$

Finally, we may expand this expression in powers of δV to write

$$\delta x \approx \frac{\epsilon}{(\epsilon F_e - \rho_e x_e)} \delta V - \frac{\rho_e \epsilon^2}{2(\epsilon F_e - \rho_e x_e)^3} \delta V^2 + \dots, \quad (11)$$

which gives

$$\frac{\delta Q}{\delta V} = \frac{A \rho_e \epsilon}{(\epsilon F_e - \rho_e x_e)} - \frac{A \rho_e^2 \epsilon^2}{2(\epsilon F_e - \rho_e x_e)^3} \delta V + \dots \quad (12)$$

Writing $|\delta Q / \delta V| = C_0 + C_1 \delta V + C_2 (\delta V)^2 + \dots$, and putting Eq. (12) in terms of the positive quantity $|\rho_e|$ so that it applies to both p - and n -type semiconductors, we have

$$C_0 = \frac{A |\rho_e| \epsilon}{(\epsilon F_e + |\rho_e| x_e)}, \quad (13)$$

$$C_1 = - \frac{A \rho_e^2 \epsilon^2}{(\epsilon F_e + |\rho_e| x_e)^3}. \quad (14)$$

The coefficient C_1 of the linear correction term thus allows us to determine the ionized gap state density at the position x_e , which is $|\rho_e|/q$:

$$N_{DL} \equiv - \frac{C_0^3}{2q \epsilon A^2 C_1} = \frac{|\rho_e|}{q}. \quad (15)$$

At this point, it is important to note that the derivation of the coefficient C_1 assumed that δV was applied *asymmetrically* with respect to the dc bias; that is, it acted to strictly *increase* the reverse bias that existed when only the dc bias was present. This is necessary to keep the charge density fixed inside of the position x_e as the amplitude of the alternating voltage is varied to determine C_1 . This points to an impor-

tant experimental requirement that must be followed in employing the DLCP method; namely, that for a given value of dc bias, the maximum voltage excursion in the forward direction must remain constant. This requires a small additional dc offset which varies with the amplitude of the ac bias. This has been noted previously⁸ and is also discussed in Sec. V. Additionally, note that the quantity δV appearing in Eqs. (11) and (12) represents the *peak-to-peak* voltage of the alternating voltage being applied.

For a *p*-type semiconductor, we will have

$$\frac{|\rho_e|}{q} = p + \int_{E_F}^{E_V + E_e} g(E, x_e) dE. \quad (16)$$

That is, the defect density measured by the DLCP method (N_{DL}) obtained experimentally from the dependence of junction capacitance on δV of the ac drive voltage, is simply related to an integral over the density of states $g(E)$ near the spatial location x_e . The integral ranges between E_F^0 (the value of the Fermi energy in the bulk), and the response depth E_e above E_V . The charge density at x_e also includes p (the net density of free hole carriers in the bulk).

Since E_e can be varied by changing the frequency or temperature of the measurement, and the position x_e can be adjusted by adding a dc reverse bias to the sample, the DLCP technique allows both the energetic distribution and the spatial uniformity of the defect population to be measured. Typically, N_{DL} is plotted against $\langle x \rangle$, where

$$\langle x \rangle = \epsilon A / C_0. \quad (17)$$

Since C_0 is the limiting value of capacitance as the ac bias amplitude goes to zero, this value of $\langle x \rangle$ is identical to that obtained from a ideal *C-V* measurement.

From Eq. (13), we see that

$$\langle x \rangle = x_e + \epsilon F_e / |\rho_e| \quad (18)$$

As long as the spatial properties of the semiconductor vary slowly, the second term on the right-hand side will be nearly constant (for a given temperature and frequency) as the dc bias is varied. This implies that differences in $\langle x \rangle$ will closely approximate actual *changes* in the position of defect response. However, it is important to keep in mind that the raw value of $\langle x \rangle$ is an average over all the contributions to capacitance, and can be shifted substantially from the position at which a predominant defect band crosses the Fermi level. Factors that can influence $\langle x \rangle$ include charged layers near the barrier interface as well as the shallow doping density if it is on the order of, or larger than, the density of deeper states. On the other hand, such issues are not expected to significantly affect the measurement accuracy of N_{DL} itself. Instead, the accurate determination of N_{DL} depends on the precise separation of the capacitance and conductance phases of the measured admittance, an accurate determination of the device area, which may depend on the roughness of the junction,¹⁹ and knowledge of the dielectric constant of the sample. The data presented here have been analyzed using $\epsilon = 11.7 \epsilon_0$. Measurements of ϵ in the literature range from $9.8 \epsilon_0$ ²⁰ to $13.6 \epsilon_0$.²¹ Other work predicts $\epsilon = (11.6 \pm 2) \epsilon_0$.²²

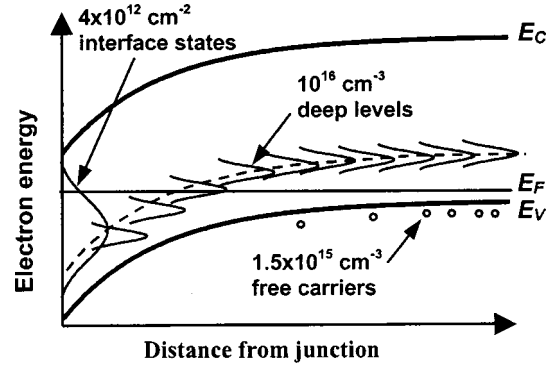


FIG. 3. Schematic of model semiconductor junction consisting of a lightly *p*-type semiconductor with a free hole density of $1.5 \times 10^{15} \text{ cm}^{-3}$, a deep acceptor defect at $E_V + 0.3 \text{ eV}$ with a density of $1 \times 10^{16} \text{ cm}^{-3}$, plus a band of interface states of density $4 \times 10^{12} \text{ cm}^{-2}$ with roughly the energy distribution indicated. The n^+ side of the junction is not shown.

IV. NUMERICAL SIMULATIONS

We have solved Poisson's equation numerically for various types of $n^+ - p$ junctions to directly demonstrate the advantages of the DLCP method over standard *C-V* profiling. Results presented here are for a model semiconducting junction in which the *p*-type film contains two spatially uniform features: a Gaussian band of deep acceptor-like defects and a shallow acceptor band giving rise to a specific density of free carriers. In this model, the device also contains a band of deep states localized near the barrier interface. These parameters, illustrated in Fig. 3(a), have been chosen to lie in a range corresponding with experimental data on actual thin-film CIGS photovoltaic junctions. It is worth mentioning that we may also easily incorporate spatial variations of the deep defect density, the shallow dopants, or both into these simulations.

Poisson's equation becomes an integral-differential equation under such conditions. The numerical solution utilizes a modified Noumerov method, which has been described in detail previously.¹⁸ For the present study, we solve the Poisson equation first at the nominal dc bias, and then add several values of δV . Each value of δV results in a change of total charge (δQ), which is found by numerically integrating over the entire depletion region. The temperature and frequency response is incorporated by imposing a thermally activated emission time limit for the gap state response. Capacitance (C_0) is then determined by computing $\delta Q / \delta V$ for a very small value of δV (typically a few mV), and C_1 is found by repeating the calculation for larger values of δV . Thus, the numerical calculation requires none of the formalism of the previous section; it simply relies on Poisson's equation and the validity of $\delta Q / \delta V$ with an emission time limit to deduce the junction capacitance.

The result of a simulated admittance spectrum based upon the model junction of Fig. 3 is displayed in Fig. 4. One clearly observes a two-peak structure in the frequency derivative of capacitance: The higher frequency peak is due to the deep bulk defect level, while the lower frequency peak comes from the states at the barrier interface. Clearly, neither feature could be identified as a bulk or interface feature from

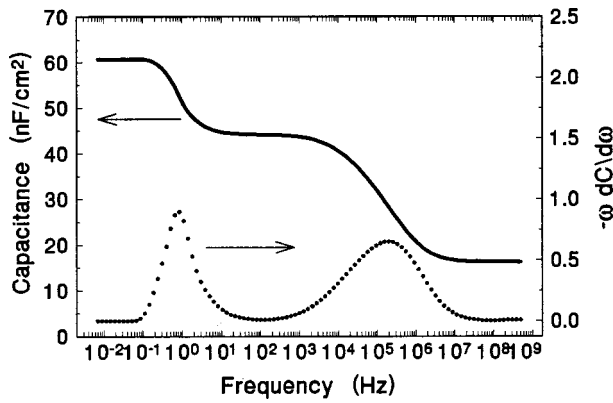


FIG. 4. Simulated admittance spectrum under zero dc bias for the model semiconductor junction of Fig. 3, assuming a built-in potential of 0.55 V. We assumed a temperature of 290 K and a thermal emission prefactor of $(7.5 \times 10^6 \text{ s}^{-1} \text{ K}^{-2}) T^2$ for both the interfacial and bulk defects.

such a spectrum by itself; however, by repeating these measurements at various applied dc biases, this issue can be resolved, at least in principle. It is thus unfortunate that almost all admittance spectra reported from studies of CIGS gener-

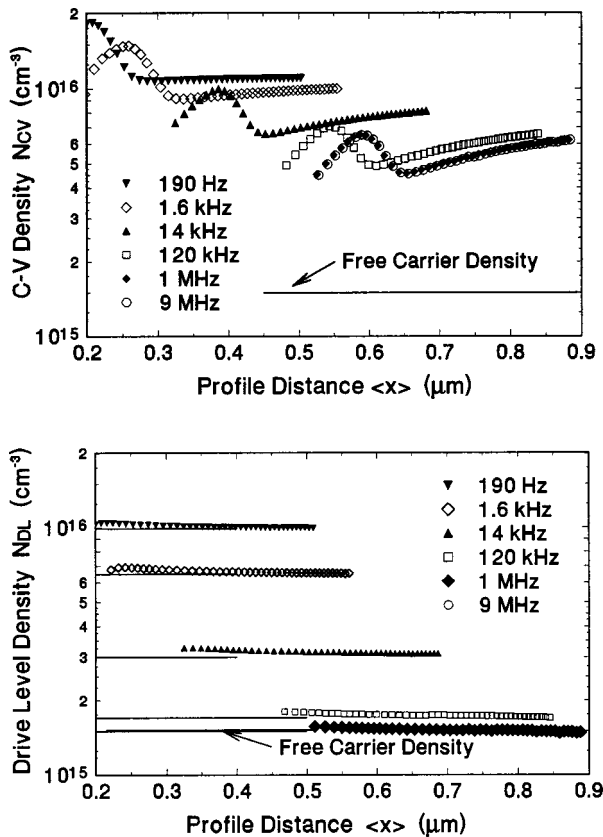


FIG. 5. (a) Simulated $C-V$ profiles for different frequencies at 290 K obtained from the model n^+-p junction shown in Fig. 3. Note that none of these profiles provide a meaningful estimate of the density of free carriers shown by the thin solid line. Also note the influence of the interface states on the shapes of these profiles. (b) DLCP profiles calculated under the same conditions. The thin solid lines indicate the theoretical value these profiles should have as given by Eq. (16). Note that in this case, the highest frequency profiles accurately reflect the density of free carriers, and that there is almost no influence from the interface states.

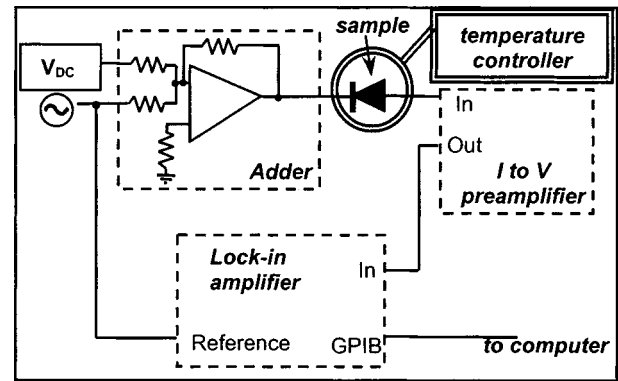


FIG. 6. Schematic of experimental setup for capacitance measurements. A capacitor is used in place of the sample for calibration of the signal phase and magnitude.

ally have only presented experimental results for zero applied bias conditions.

A set of simulated $C-V$ profiles for this model semiconductor junction is displayed in Fig. 5(a). This simulation clearly demonstrates that both the deep states and the interface states have a profound influence on the $C-V$ profiles. In particular, their influence almost completely precludes an accurate measurement of the free carrier density by this method. In Fig. 5(b), we display the simulation of a DLCP measurement. As expected, the profile for the largest values of E_e (smallest frequency) most closely agrees with the $C-V$ results, and accurately reflects the maximum density of states that can respond to the alternating component of bias: deep states plus shallow acceptors. However, in the low E_e limit (highest frequencies), the DLCP measurement discloses a much smaller density than $C-V$, and this accurately reflects the density of free carriers. This is because DLCP is a purely dynamic measurement so that, while the nonuniform, static charge density created by the ionized deep defects influence the value of the depletion width (W), it does not affect the value of N_{DL} . For the same reason, DLCP is insensitive to interface states, which clearly influences the $C-V$ results shown in Fig. 5(a). This attribute of DLCP has previously been discussed in some detail.⁸

V. EXPERIMENTAL DETAILS

A. Instrumentation for DLCP

The equipment required for a DLCP measurement is identical to that for admittance spectroscopy or $C-V$ measurements. Our setup is illustrated in Fig. 6. Forming the core of this measurement are the Stanford Research Systems (SRS) digital lock-in amplifier (model SRS850) and the SRS current-to-voltage preamplifier (model SRS570). This measurement could also be done with a standard LCR meter, assuming the amplitude of the applied ac voltage was adjustable. The device temperature is adjusted by heating or cooling through a N_2 gas flow cryostat, and temperature is measured with a thermocouple. This system, with the use of liquid nitrogen cooling, allows adjustment of the temperature between approximately 90 and 400 K with accuracy of approximately ± 1 K.

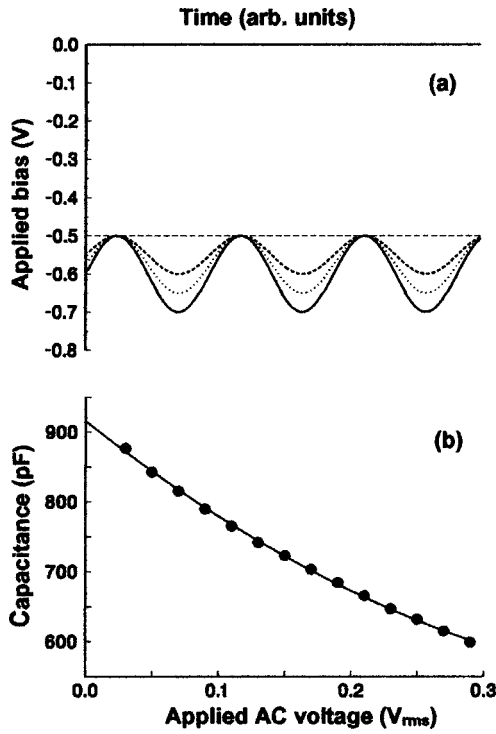


FIG. 7. (a) Illustration of the manner in which ac and dc bias must be changed in tandem, so that the position x_e remains the same throughout the DLCP measurement. (b) Typical raw DLCP data showing the variation of junction capacitance with ac bias amplitude. The nominal reverse dc bias was 0.1 V. The solid line shows a quadratic fit, yielding coefficients $C_0 = 912$ pF, $C_1 = 509$ pF/V, and $C_2 = 152$ pF/V². Since the device area was 0.03 cm², this measurement results in values of $N_{DL} = 5 \times 10^{15}$ cm.

There is only one nonobvious aspect to this measurement. This was alluded to in Sec. III; namely, as the amplitude of the ac voltage (V_{ac}) is adjusted over the course of the experiment, the dc bias (V_{dc}) must simultaneously be adjusted so that the maximum applied voltage $V = V_{ac} + V_{dc}$ stays constant, and thus x_e remains fixed.⁸ This situation is illustrated in Fig. 7(a).

For a given ω , T , and V_{dc} , capacitance is usually measured for 10 to 15 values of V_{ac} , typically from $0.01 V_{rms}$ to as much as $0.3 V_{rms}$. A typical set of raw C versus V_{ac} data is shown in Fig. 7(b). A quadratic fit to the data allows values of C_0 and C_1 to be extracted, from which Eq. (15) yields N_{DL} . The measurement is then repeated for different values of dc (reverse) bias, creating a spatial profile of N_{DL} for a fixed value of E_e . Finally, ω or T is varied and the entire

process repeated to show the energetic dependence of the profile. It is somewhat preferable to vary E_e by changing ω at fixed T , since varying T could result in a shift in E_F^0 and thus make the interpretation of the results more difficult. However, varying T allows a much wider range of E_e values to be accessed, and this is often necessary to examine the full range of the gap state response.

B. Sample preparation

Devices for this study were fabricated at the Institute of Energy Conversion (IEC). The CIGS films were grown by four-source evaporation on Mo-coated glass substrates, and are about $2 \mu\text{m}$ thick. The films discussed here were prepared using a substrate temperature of 550°C , resulting in grain sizes averaging around $1\text{--}2 \mu\text{m}$. CdS was deposited to a thickness of $30\text{--}50$ nm by chemical bath deposition. Finally, a 50 nm thick layer of ZnO:Al with resistivity $\rho = 50 \Omega\text{--cm}$ was deposited by rf scattering followed by either a 500 nm thick ZnO layer with sheet resistance $20 \Omega/\text{sq}$ (samples D008, D934, D912) or a 170 nm thick ITO layer with sheet resistance $25 \Omega/\text{sq}$. Device preparation is discussed more completely elsewhere.²³ Properties of some representative devices are summarized in Table I.

C. Light soaking

In addition to measuring the density of states in the gap for these films, we wished to examine whether this bulk density of states could be altered by optical exposure. Prior studies of optically induced metastabilities in these films have been interpreted to infer changes occurring primarily in the near-interface regions in CIGS devices (see, e.g., Ref. 3). Indeed, the optical excitation employed in those studies always included visible wavelengths, which are strongly absorbed by the CIGS film and thus do not penetrate very far into the film. Therefore, in order to explore whether bulk defect densities could be perturbed by optical excitation independently of interface effects, we employed the 1064 nm line of a cw YAG laser, which would be more uniformly absorbed throughout the CIGS film. For such studies, samples were exposed to approximately $100 \text{ mW}/\text{cm}^2$ for 3 h at 300 K, then immediately cooled in the dark to below 250 K.

TABLE I. Details of representative devices. Properties listed include bandgap (E_g) and Ga fraction [Ga/(In+Ga)]. Also listed are performance characteristics: efficiency, open circuit voltage (V_{oc}), short circuit current (J_{sc}), and fill factor (FF). Results from DLCP measurements include estimates of the free carrier density and deep defect density. Ranges result from a spatially nonuniform response.

Sample #	E_g (eV)	Ga fraction	Eff (%)	V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	$p_{(est)}$ (10^{15} cm^{-3})	$N_{D(est)}$ (10^{15} cm^{-3})
D469	1.0	0.0	11.3	0.46	36.8	66.6	6.5	3.5–8.5
D008	1.0	0.0	9.53	0.41	65.1	65.6	0.7–3	5.3–17
D456	1.16	0.27	11.6	0.55	32.4	65.0	9–19	21
D934	1.18	0.30	14.9	0.63	30.6	76.8	2–6	1
D400	1.20	0.32	16.1	0.65	33.0	74.8	3–5	2–3
D912	1.53	0.80	8.8	0.82	16.3	65.9	0.6	1.4

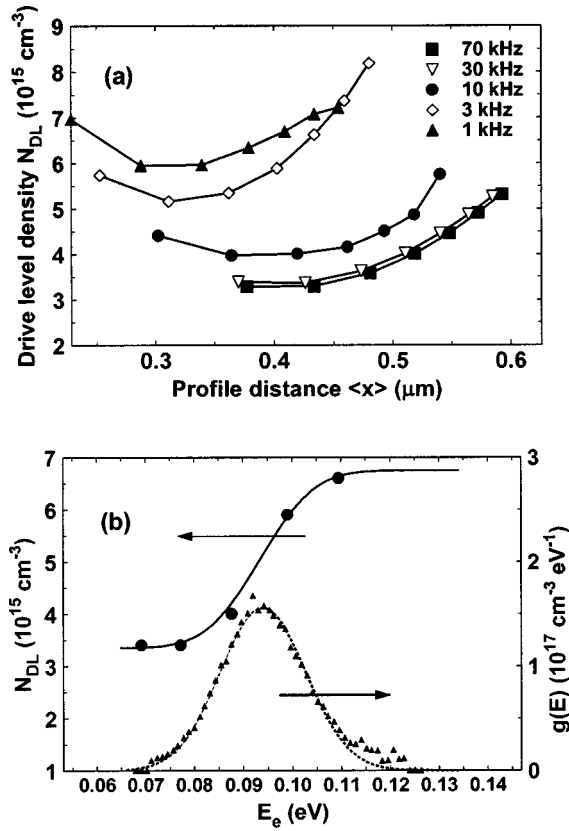


FIG. 8. (a) Typical DLCP profiles, taken for sample D400 at 110 K using several frequencies, as noted in the figure. (b) Plot of the DLCP value at $\langle x \rangle = 0.4 \mu\text{m}$ versus thermal energy E_e , showing more clearly the thermal activation of the defect. E_e was calculated using Eq. (1), with $\nu = 5.4 \times 10^4 \text{ T}^2 \text{ s}^{-1}$ obtained from admittance spectroscopy. The fit to the DLCP response, shown by the solid line, assumes a Gaussian shape which corresponds in position and width to the admittance spectroscopy data ($-\omega dC/d\omega$, triangles, scaled in magnitude to match the DLCP result).

VI. EXPERIMENTAL RESULTS

A. General characteristics of DLCP profiles for CIGS

A series of DLCP profiles taken at a series of different frequencies at fixed temperature (110 K) is displayed in Fig. 8(a) for a $\text{CuIn}_{1-x}\text{Ga}_x\text{Se}$ device with $x=0.3$. We see that, while there is a significant spatial dependence to the electronic properties for this sample, they are sufficiently slowly varying so that, with reasonable confidence, we may extract the integrated density of states versus emission energy, $E_e = k_B T \log(\nu/\omega)$, in the vicinity of a particular spatial location, say, where $\langle x \rangle$ is near $0.4 \mu\text{m}$. These extracted data are displayed by solid circles in Fig. 8(b). Comparing these data with Eq. (16) suggests that the free carrier density (obtained in the low E_e limit) is roughly $3.5 \times 10^{15} \text{ cm}^{-3}$, and that there is a deeper defect with a thermal emission energy near 0.1 eV of density $3.3 \times 10^{15} \text{ cm}^{-3}$. This is seen more clearly by the dotted curve in Fig. 8(b), which represents the derivative of the solid line with respect to E_e ; that is, the density of states. Finally, the triangles plotted on top of the dotted curve indicate the dependence of $\omega dC/d\omega$ obtained from admittance spectroscopy data of this sample under 0 V dc bias. These data have been rescaled to match the magnitude of the dotted curve. The peak activation energy and width of the

defect band obtained from these two measurements agree quite closely. However, considerably more information about the electronic properties of the sample has been deduced from DLCP, specifically, we can assess the spatial uniformity of the deduced defect distribution, and we obtain a direct quantitative value for the defect's density. Indeed, in samples in which the spatial variation of properties is more pronounced, the use of admittance spectroscopy to deduce the density of states can lead to large errors.

The quantitative accuracy of the DLCP measurement was previously demonstrated for the D_0 center in amorphous silicon films by comparing results using electron spin resonance (ESR) measurements with DLCP in $a\text{-Si:H/Pd}$ Schottky devices.¹⁷ Since DLCP can only examine deep states between E_F and E_e , and because the D_0 center lies very near midgap in $a\text{-Si:H}$, it was recognized that the DLCP measurement was probably not capable of recording the entire defect response from that center. For this reason, in Ref. 17, the total defect density was estimated to be *twice* the measured DLCP defect density determined at the position of maximum slope in the N_{DL} versus E_e data. Following this procedure, excellent agreement was obtained between ESR and DLCP estimates for the D_0 density in $a\text{-Si:H}$, to within a factor of 1.3. Unlike the case of $a\text{-Si:H}$ films, the predominant deep defect response in CIGS films is much shallower than midgap, and we believe that DLCP is able to detect the entire band. Thus, the defect densities deduced from DLCP for our CIGS samples are taken to be the *difference* between the saturated maximum N_{DL} values, at high E_e , and the estimated free carrier density, obtained from the low E_e limit of N_{DL} (see Table I).

The spatial sensitivity of the DLCP measurement has also been tested in $a\text{-Si:H}$ films, most notably for samples with an intentional spatially varying concentration of carbon impurities with a modulation period of $0.4 \mu\text{m}$.¹⁶ In this study the secondary-ion mass spectroscopy (SIMS) profiles of these impurities were compared to the spatial profiles obtained from DLCP. A distinct spatial variation was indeed evident in the DLCP profiles, and the distances between positions of peak impurity concentration corresponded to those measured by SIMS. However, the relatively abrupt changes in concentration evident in the SIMS profiles appeared smeared over a distance of about $0.1 \mu\text{m}$ in the DLCP profiles. This reflects the limits on spatial resolution in the DLCP measurement, corresponding roughly to the Debye length.

As a second example we display, in Fig. 9, a direct comparison between the standard $C-V$ profiles with the DLCP profiles for a sample containing 0% Ga (that is, for a CuInSe_2 device). In this case, we have varied E_e by examining profiles at fixed frequency (11 kHz), but over a range of measurement temperatures. We estimate that, for this sample, the lowest temperature profile at 190 K corresponds to a value of E_e of 0.25 eV, while the highest temperature (280 K) corresponds to about 0.36 eV. Admittance spectroscopy for this device indicates a deep defect with a characteristic energy near 0.3 eV; thus, our range of profiles spans the energy response of this center. Indeed, we see that both the high- and low-temperature DLCP profiles in Fig. 9(b) each

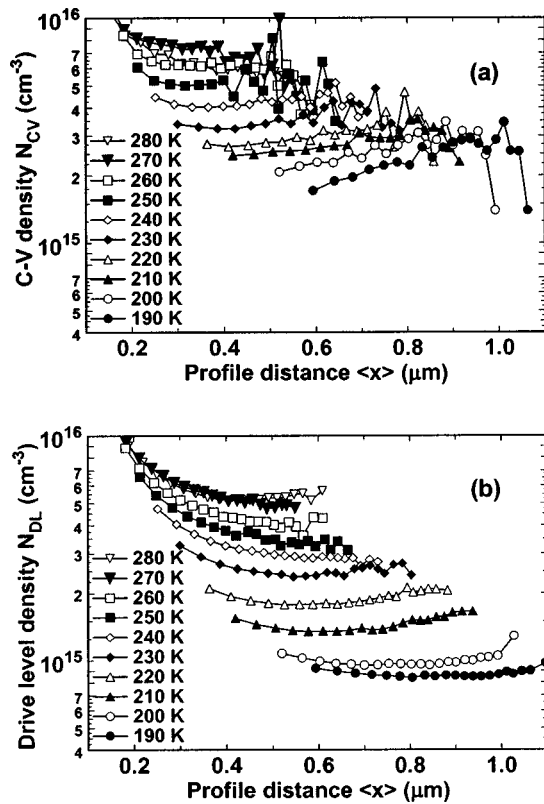


FIG. 9. Comparing (a) $C-V$ and (b) DLCP results. These data were taken for device D008 using 11 kHz and the series of temperatures noted in the figure.

appear to reach a limiting value. The low-temperature value of slightly below 10^{15} cm^{-3} thus indicates our best estimate for the free carrier density, while the difference between the high- and low-temperature DLCP profiles (of roughly $5 \times 10^{15} \text{ cm}^{-3}$) gives us the density of the deep defect itself.

Such estimates are much more difficult based upon the $C-V$ profiles of Fig. 9(a). Even at the lowest temperatures, we would overestimate the free carrier density by more than a factor of 2 and also judge it to be somewhat nonuniform. Both of these conclusions would be false based upon the DLCP profiles. In addition, the uniformity of the deep defect response indicated for the highest temperature profiles differ substantially for the $C-V$ versus DLCP results: DLCP indicates a definite increase as we move within $0.4 \mu\text{m}$ of the barrier junction. It is actually quite interesting to compare these two sets of profiles with the simulated profiles appearing in Fig. 5 since many aspects of that simulation are mirrored in these actual data. Note, however, the $C-V$ profiles do not indicate a specific feature at low applied bias, resembling the portion of the simulated response [Fig. 5(a)] that was due to an assumed interface state.

DLCP data have been collected for a number of devices. Some representative results are given in Table I. When a device exhibited nonuniformity, a range of results is indicated. Clearly this is not an uncommon occurrence. We have also observed no correlation between the density of this defect and the device performance. Perhaps this is unsurprising, since it is a relatively shallow defect and so may not act as a significant recombination center. Indeed, other comple-

mentary measurements, which are sensitive to a range of energies not accessible by DLCP, have detected defects which may be more significant for the device performance.²⁴

Finally we should mention that, for all devices studied, the admittance spectra have indicated an activation energy of conduction well below 100 meV. This means that conduction cannot be frozen out using the range of temperatures and frequencies currently available to us, and so we can only set an upper limit on the value of the actual free carrier density.

B. Bulk versus interface nature of the deep defects

We next wish to use such DLCP profiles to address the issue of interface states. First of all, the deep defect feature appearing in the admittance spectra for the samples in either Figs. 8 or 9 can definitely be identified as originating from defects in the bulk absorber layer. In these data, DLCP actually provides us with its spatial profile over a distance exceeding $0.3 \mu\text{m}$ in each film. Generally all of our admittance spectra on these samples give clear evidence for a deep defect response that exhibits an activation energy varying between 0.1 and 0.3 eV, depending on the sample. In addition for all the samples studied, the thermal emission energies and prefactors of the dominant deep defect were related according to the Meyer-Neldel rule.^{25,26} We believe this defect response for our series of samples matches those referred to as N_1 commonly reported in previous studies of CIGS samples using admittance spectroscopy. That is, the deep defect response exhibited by our samples occurs within the same range of energies as those labeled N_1 and follows the same Meyer-Neldel variation.^{27,28} We mention in passing that devices deposited at lower substrate temperatures exhibited an additional, very broad defect response with a characteristic energy close to midgap (around 0.55 eV) and an emission prefactor $\nu_0 = 4 \times 10^7 \text{ cm}^{-1} \text{ K}^{-2}$. Some additional discussion of this feature is given in Ref. 29.

As with the two films examined previously we have been able to use DLCP to profile the N_1 response through each of our films. While we have found some devices to be more spatially uniform than others, these results unambiguously demonstrate that the N_1 defects we observe must originate from the bulk region of the CIGS film and not from defects near the CIGS/CdS interface. Indeed, the DLCP measurement would generally be quite insensitive to any response from interface states [see Fig. 5(b)].

Previously, some scientists have argued that N_1 must be associated with a defect near the barrier interface and, more specifically, a minority carrier trap whose admittance response was due to electron carrier trapping coming from the CdS side of the junction.³⁰ This argument was based in part upon the fact that the N_1 response could be altered using surface treatments that were believed to only affect a region near the top contact of sample devices. We note that if the N_1 response were indeed due to a minority carrier trap at the interface whose electronic occupation was altered by the exchange of electrons with the n -side of the junction, then changes in capacitance observed due to such a process would only reflect changes in the depletion region response on that side of the junction. This depletion region at most may

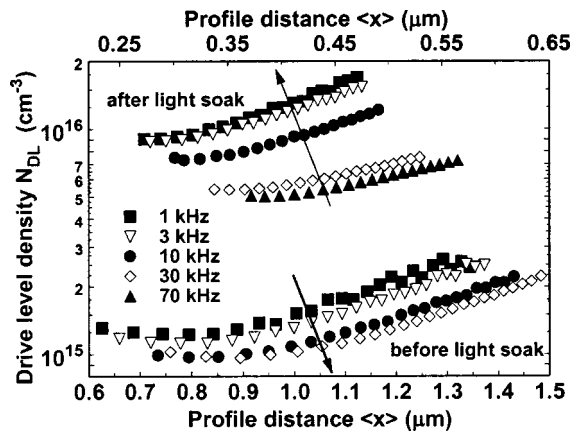


FIG. 10. DLCP profiles, before and after light soaking, taken at 135 K. Note that the x axes differ for the two sets of data. These profiles clearly illustrate a significant increase in carrier density (assumed to dominate the response at high frequencies) and also an increase in the deep trap density. This device was prepared using $T_{ss}=400^\circ\text{C}$, but the metastable behavior observed is characteristic of all devices we have studied.

barely extend into the ZnO film, which would limit the largest possible capacitance step observed in the admittance spectra to a value slightly larger than the total width of the CdS layer. Given that this layer is generally no more than 30–50 nm thick, and that the N_1 feature typically indicates a change of depletion capacitance corresponding to at least 0.2 μm , and sometimes is even larger, there is no possibility that electron trapping from the CdS side of the junction could be responsible for this feature.

C. Metastable effects due to light soaking

After concluding that N_1 must be a bulk defect in the CIGS absorber itself, we wished to find out whether its properties could be perturbed by optical exposure. Using the optical treatment at long wavelength discussed previously, we have found that both the bulk carrier density in the absorber and the N_1 defect density are significantly changed, as illustrated in Fig. 10. Our measurements gave no indication that interface states were introduced or increased by this light-soaking method. On the other hand, some kinds of interface states may only be visible under a fairly narrow range of experimental conditions, so that we cannot completely rule out additional metastable effects at the interface. We also determined the photovoltaic device efficiency of this sample before and after our light treatment using a dichroic filtered tungsten halide (ELH) light source at 100 mW/cm² intensity. We found a commensurate reduction in conversion efficiency, primarily due to a change in fill factor, from roughly 13% to 11%. In all devices studied, the metastable changes we observe anneal away in the dark at elevated temperatures, with a time scale of several hours at 50 $^\circ\text{C}$.

Previous experiments by other groups employing prolonged optical exposure have noted similar increases in depletion charge density, and possibly carrier density, indicated by an increased junction capacitance.^{2,3} Our results indicate that an important component of such changes may actually come from the bulk film. We are also able to clearly distinguish changes in carrier density from changes in the

density of deep defects. We can suggest several possible scenarios for how the observed metastable effect might come about. (1) It indicates a real increase in the shallow dopant and deep trap densities, possibly through linked precursor sites that are activated by photoexcitation. (2) The capture of photogenerated minority carriers stabilizes a different charge configuration of the defect involved. This inhibits the release of the trapped electrons until the sample is heated. The increase of the deep defect and shallow acceptor density we observe then belong to two different charge states of that metastable defect site.³¹ (3) Additional defects are created due to the effects of mobile Cu^+ ions which respond to changes in the field (depletion region collapse) caused by excess photogenerated carriers.^{4,32}

Most of these explanations imply a quantitative link between changes in the deep defect density and the density of shallow dopants. Indeed, the data in Fig. 10 seem to indicate that the changes in deep defect density and the changes in carrier density are nearly equal (roughly $4 \times 10^{15} \text{ cm}^{-3}$ are added to both values). However, the spatial nonuniformity of the devices studied to date makes such quantitative statements ambiguous at best. The net shallow dopant density and deep defect density, could, with similar confidence, be interpreted as increasing by a similar *factor* (approximately a factor of five to ten) depending on the spatial alignment we use to make such comparisons. Clearly, the physical mechanism we might infer from the observed metastable changes would depend crucially upon any such detailed relationship between changes in the carrier concentrations and the deep defect density. Without the spatial profiling capability of DLCP, however, we would not even have been aware of this difficulty. Hopefully, we will be able to resolve such ambiguities with additional studies in the near future, possibly by finding samples that exhibit a greater degree of spatial uniformity.

VII. CONCLUSIONS

Using the drive-level capacitance profiling (DLCP) technique, we have obtained information about the deep defect distributions in CIGS films, which is not easily deduced from admittance spectroscopy or $C-V$ measurements. In particular, we have been able to identify a defect response, with a characteristic energy around 0.1–0.3 eV from the valence band edge, as a feature within the bulk region of the CIGS film. This response appears to correspond to the N_1 defect, which has frequently been observed in these films. Although this is at face value a fairly simple observation, it has important implications for our understanding of the electronic properties of CIGS. Since the initial suggestion that N_1 originates from a continuous density of states at the interface, numerous authors have used this response to infer the position of the electron quasi-Fermi energy at the interface. From this inference, further conclusions about the electronic properties of devices have then been made [see, e.g., Refs. 33–35]. However, since DLCP indicates that in our devices this defect exists throughout the bulk region CIGS film, we now know that its activation energy does not necessarily simply correspond to the position of the electron quasi-Fermi energy at the interface.

In addition to distinguishing the bulk from the near-interface defect response, the DLCP measurement allows us to more accurately deduce the carrier density. We have found that, in some films, the carrier density is much lower than that indicated by C - V measurements due to the response from deeper states. The DLCP technique also provides a spatial profile of the film, allowing the degree of spatial nonuniformity in the density of defect states to be observed.

Finally, metastable effects in CIGS photovoltaic devices are well known, but their origin is still a mystery. Changes in both the interfacial and bulk absorber properties have been discussed.^{35,36} Distinguishing between metastable changes at the junction interface vs. changes to the bulk semiconductor, is a crucial first step toward understanding the effects of treatments such as light soaking, or annealing the films in air. We have shown how DLCP provides an important tool for making such a distinction. In particular, we have demonstrated that light soaking with uniformly absorbed light results in changes to both the carrier density and defect density in the bulk film. Ultimately, we hope that the quantitative relationship between the metastable changes in carrier densities and deep defect densities revealed by DLCP will help us identify the type of underlying mechanism responsible for such metastable effects.

The results from the DLCP measurements reported here immediately suggest some additional studies that may enhance our fundamental understanding of CIGS films. For example, the bandgap of films is usually intentionally graded during growth, by variation of the elemental composition, to enhance the photovoltaic device performance. The DLCP method provides the most reliable means to spatially profile the resulting electronic properties within these films. Thus, one should be able to establish a quite detailed relation between the composition and the electronic properties within a single sample. This would thus allow the effects of quite small changes in film composition on their properties to be documented. Alternatively, one could study identically grown films made into devices with their n^+ -junction layers fabricated in different manners. Such different types of junctions would be expected to affect the interfacial properties; however, due to the metastable behavior of CIGS, this could also alter the material quite some distance into the bulk film. We believe that the DLCP method, coupled with admittance spectroscopy, would be able to distinguish these two possibilities.

Clearly, such types of studies could have an enormous impact on the efforts to understand CIGS photovoltaic devices. Efforts to model such devices mathematically have grown substantially in the past year.³⁷ We believe that the energetic and spatial distributions of electronic states determined experimentally by DLCP will serve as critical input toward helping these efforts succeed.

VIII. ACKNOWLEDGMENTS

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